Jared Admiraal

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Education

University of California, San Diego La Jolla, CA
B.S. Aerospace Engineering (Flight Dynamics & Controls), GPA: 3.92 *Expected June 2027*

- Relevant Coursework: Graduate Fluid Mechanics, Numerical Methods, CFD, FEM, Graduate Applied Mathematics (PDEs, Asymptotics), Controls, Dynamics, Advanced Vibrations, Propulsion, Aerodynamics, Structures
- Activities: Triton Racing Formula SAE (Aerodynamics Team), Instructional Assistant (PDEs)

Experience

Flight Sciences Intern Jun 2026 – Present
ESAero (Empirical Systems Aerospace) *San Luis Obispo, CA*

- Supporting flight sciences for an uncrewed aircraft program spanning mission analysis, CFD, aerodynamics, and guidance, navigation, and control (GNC).

Researcher - Computational Cellular Mechanobiology Lab Jul 2024 – Present
University of California, San Diego *La Jolla, CA*

- Built a reproducible FEniCSx pipeline predicting 3D skeletal-muscle fiber orientation by solving incompressible Stokes flow (Taylor-Hood mixed FEM, Lagrange-multiplier slip) on anatomical meshes; robustness across three muscle geometries agreed within a 2.51° mean deviation.
- Reformulated fiber prediction as a Frank-Oseen liquid-crystal director field minimizing elastic energy under a unit-norm constraint, solved by projected gradient descent; reconstructed a left-ventricle cardiac fiber field and benchmarked it against the Stokes-flow model.
- Derived reduced-order, slow-manifold models for curved annular soap films with soluble surfactant via long-wave asymptotics and symbolic computer algebra, extending classical flat-film theory to curved geometry.
- Validated predictions against diffusion tensor imaging, ran boundary-condition sensitivity analyses, and was selected to present a research poster at the Biophysical Society Annual Meeting (Feb. 2026).

Aerodynamics Engineer - Formula SAE Sep 2025 – Present
Triton Racing, UC San Diego *La Jolla, CA*

- Led front and rear wing design from multi-element airfoil selection through a CFD design-of-experiments optimization; validated the full car with a grid-independent Ansys GEKO study (179 N downforce, $C_L = 2.72$) across pitch, heave, yaw, roll, and rotating-frame skidpad sweeps. Selected as Aerodynamics and Vehicle Dynamics Lead for the 2026–27 season.
- Ran a 9% scale wind-tunnel campaign that validated the CFD trends to within 4% on the normal-to-axial force ratio, confirming dynamic-pressure scaling ($R^2 > 0.996$).
- Sized composite spars and topology-optimized aluminum mounts in FEA (Puck criterion); produced GD&T drawings and manufactured the package in-house with CNC-milled molds and a quantified vacuum wet-layup process.

Systems Analysis Engineering Intern Jun 2025 – Sep 2025
AT2 Aerospace *Santa Clarita, CA*

- Created a MATLAB-based hybrid airship mission simulator from first principles, adhering to aerospace performance and operational constraints for mission feasibility analysis.
- Developed physics models for buoyancy, atmosphere, aerodynamics, propulsion, and fuel consumption to enable rapid iteration across vehicle configurations during early-stage design.
- Performed mission-level parameter sweeps to generate performance envelopes and evaluate payload, range, and feasibility under realistic flight conditions using FAA-referenced assumptions.

Technical Skills

Modeling & Simulation: PDE-based modeling, finite element methods, CFD, continuum mechanics
Scientific Computing: Python, MATLAB, computer algebra, Java, JavaScript, HTML
Libraries: FEniCSx, fTetWild, PETSc, NumPy, SciPy, Pandas, matplotlib, PyTorch, meshio
CAD/CAE: Ansys, SolidWorks, FEA, Fusion CAM, GD&T
Systems & Software Tools: Docker, Git, Embedded Linux, ROS2, LaTeX, Excel
Manufacturing & Prototyping: CNC & manual machining, composites, laser cutting, 3D printing, waterjetting, hot-wire foam cutting, photolithography, soldering
Languages: English (Native), Mandarin Chinese (Conversational), Dutch (Conversational)